data meetup

Microsoft Data Platform Meetup Groningen

2026-04-16

Mastering Power BI Model Types

Import-, DirectQuery- or Composite models?

Who am I

Or who makes that noise…

- Gregor Prohaska
- Operations Manager @ COSMO CONSULT
- „Some“ years of experience

@gregorprohaska.bsky.social

- Father of 2
- Curious by nature
- Don’t take myself to seriously ;-)

AGENDA

01 Terms and Concept

02 Single-Mode Models

03 Composite Models

04 Which one to use...

Terms and Concepts

Source Groups & Relationships

Analysis Services engine

Storage Modes

Semantic Model

Performance

Analysis Service Engine

- Query-Execution Engine
- Formula Engine (FE) generates query plan and solves complex tasks
- Storage Engine (SE) retrieves data from data store

Further reading: https://www.sqlbi.com/articles/formula-engine-and-storage-engine-in-dax/

Terms and Concepts

Source Groups & Relationships

Storage Modes

Semantic Model

Performance

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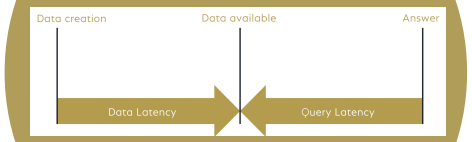
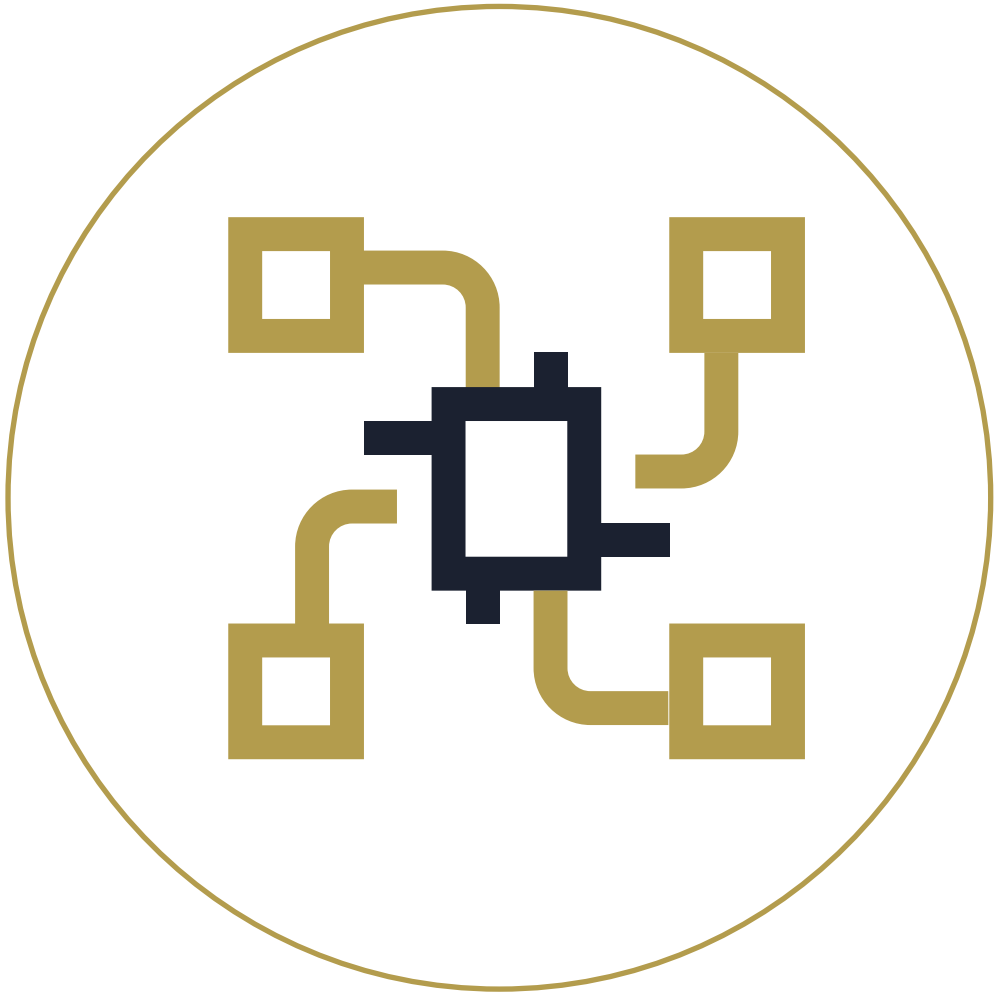
Storage Modes

Semantic Model



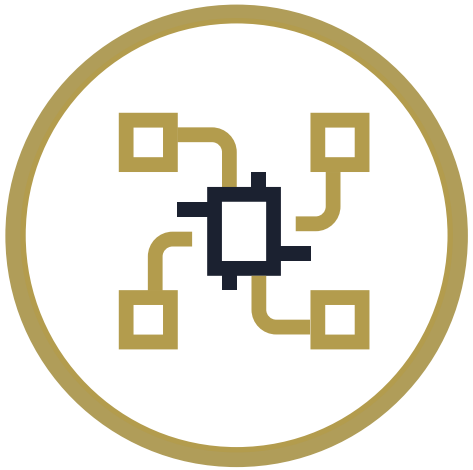
Semantic Model

- Logical layer
- Defines tables & their relationships
- Provides metadata such as formating & aggregations

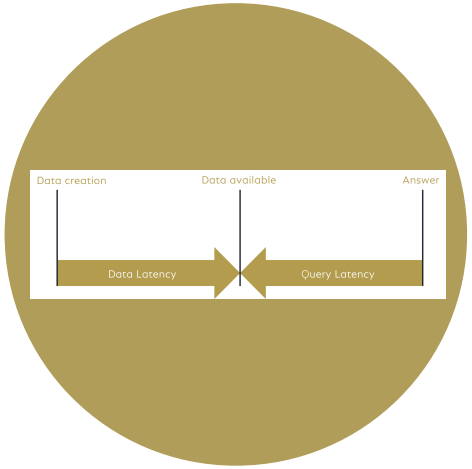


Terms and Concepts

Source Groups & Relationships

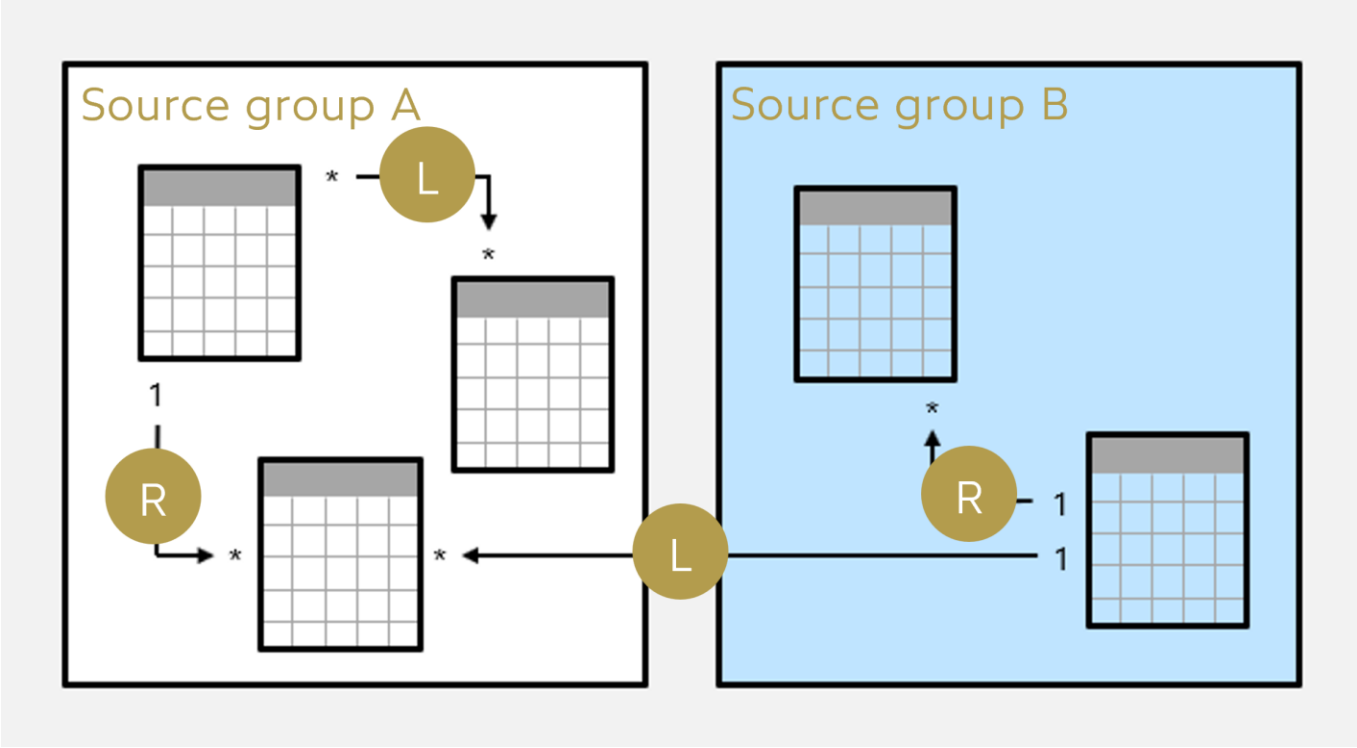


Storage Modes



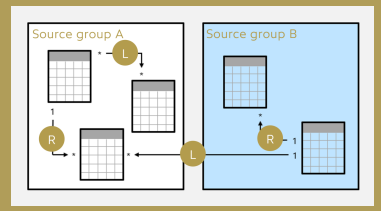
Source Groups & Relationships

- Each Source is a source group
- Regular relationships are created for 1-N relations within a source group
- Limited relationships for M-N or cross source group relations

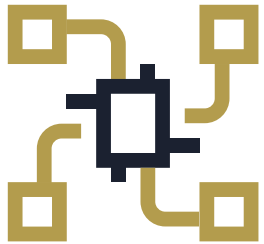


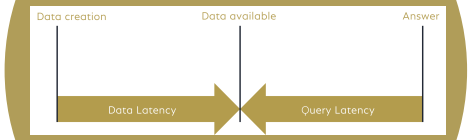
adapted from https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-relationships-understand#relationship-evaluation

Terms and Concepts



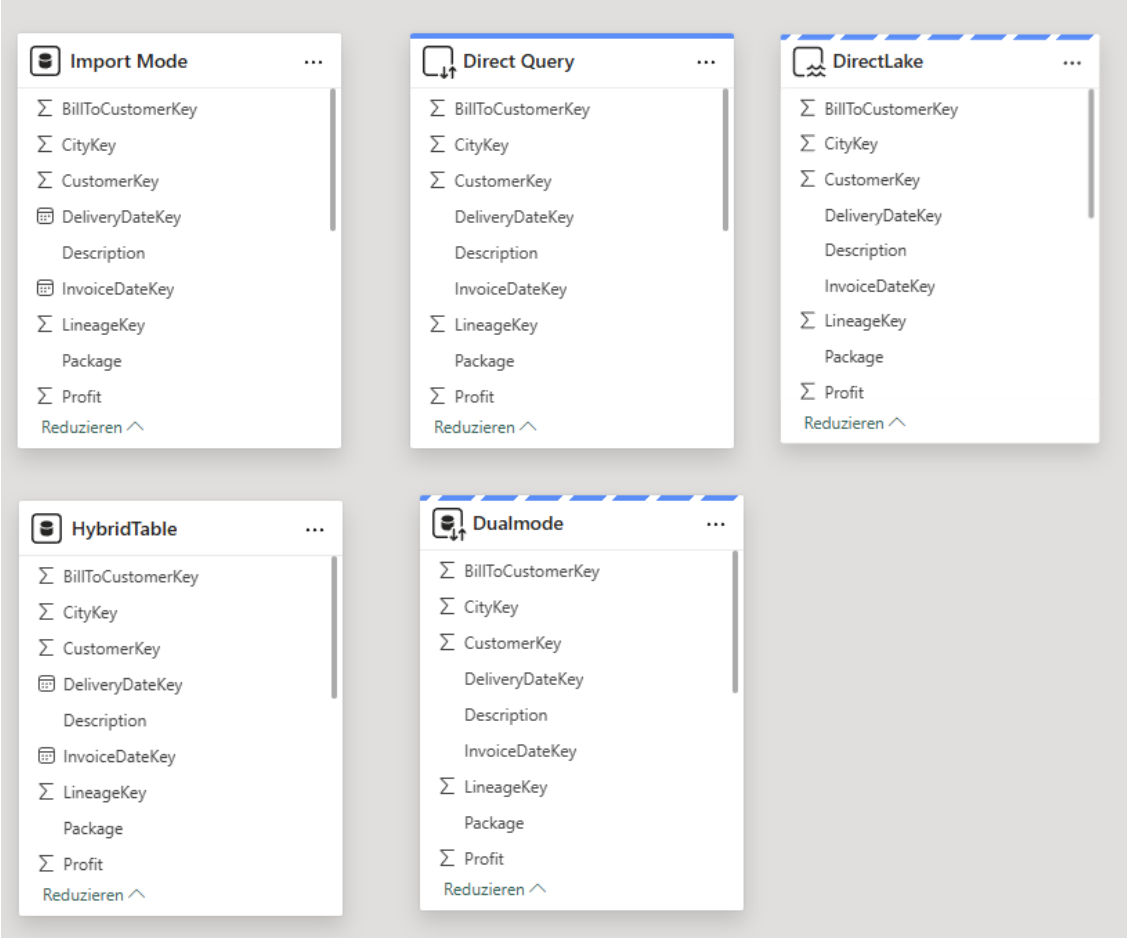
Storage Modes





Storage Types

- Defined per table
- three main types
- two variations



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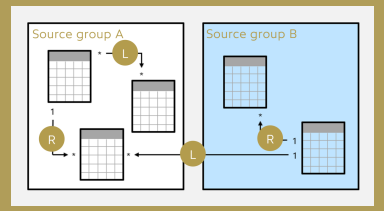
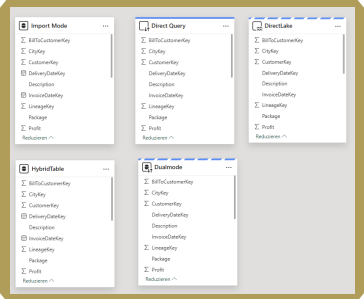
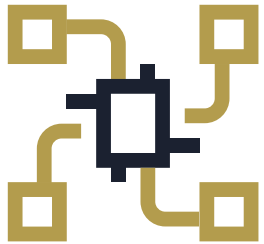
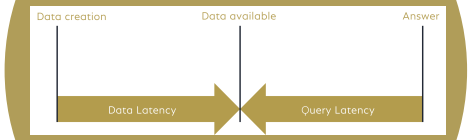




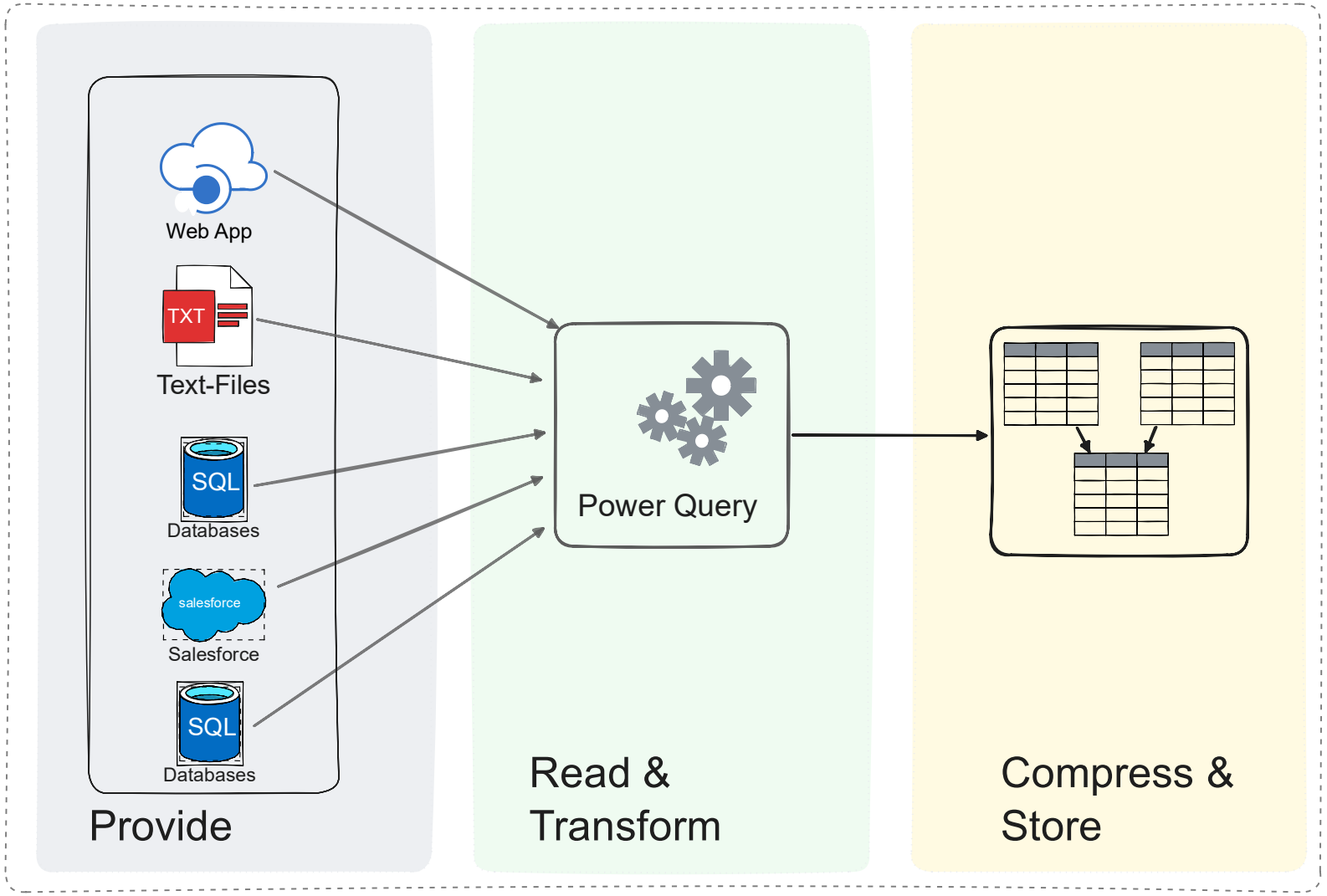




Single-Mode Models

1. Import Mode
2. DirectQuery Mode
3. Direct Lake
   - on OneLake
   - on SQL Analytics Endpoint

Import Model



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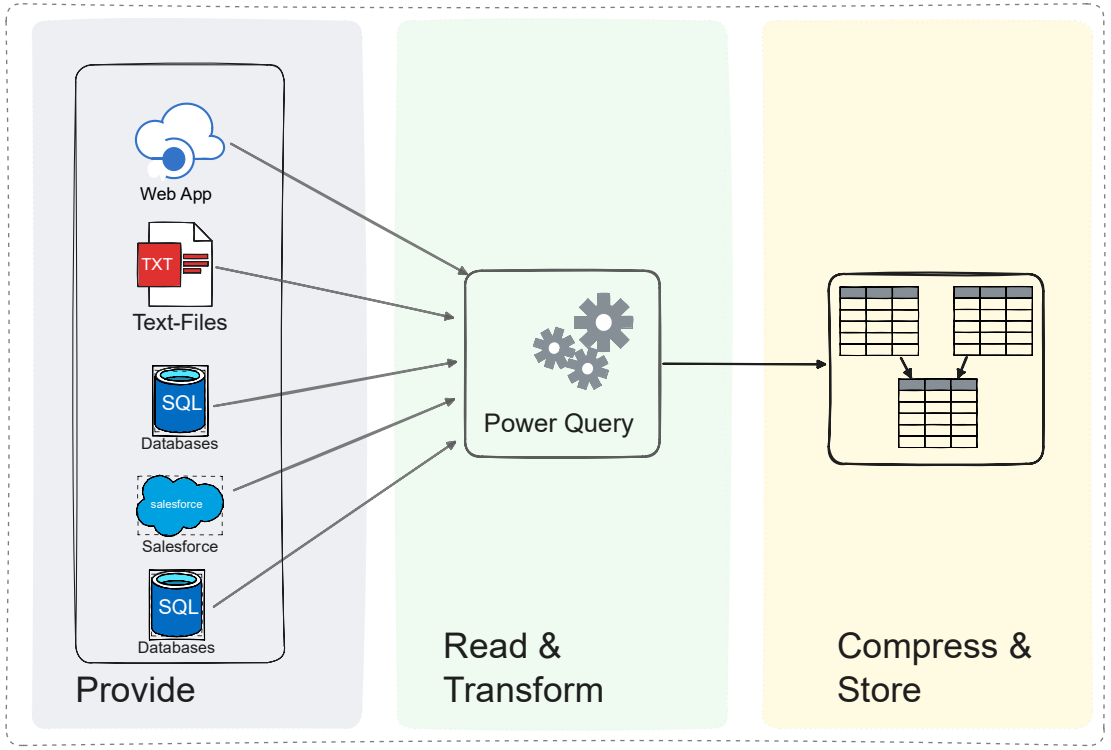
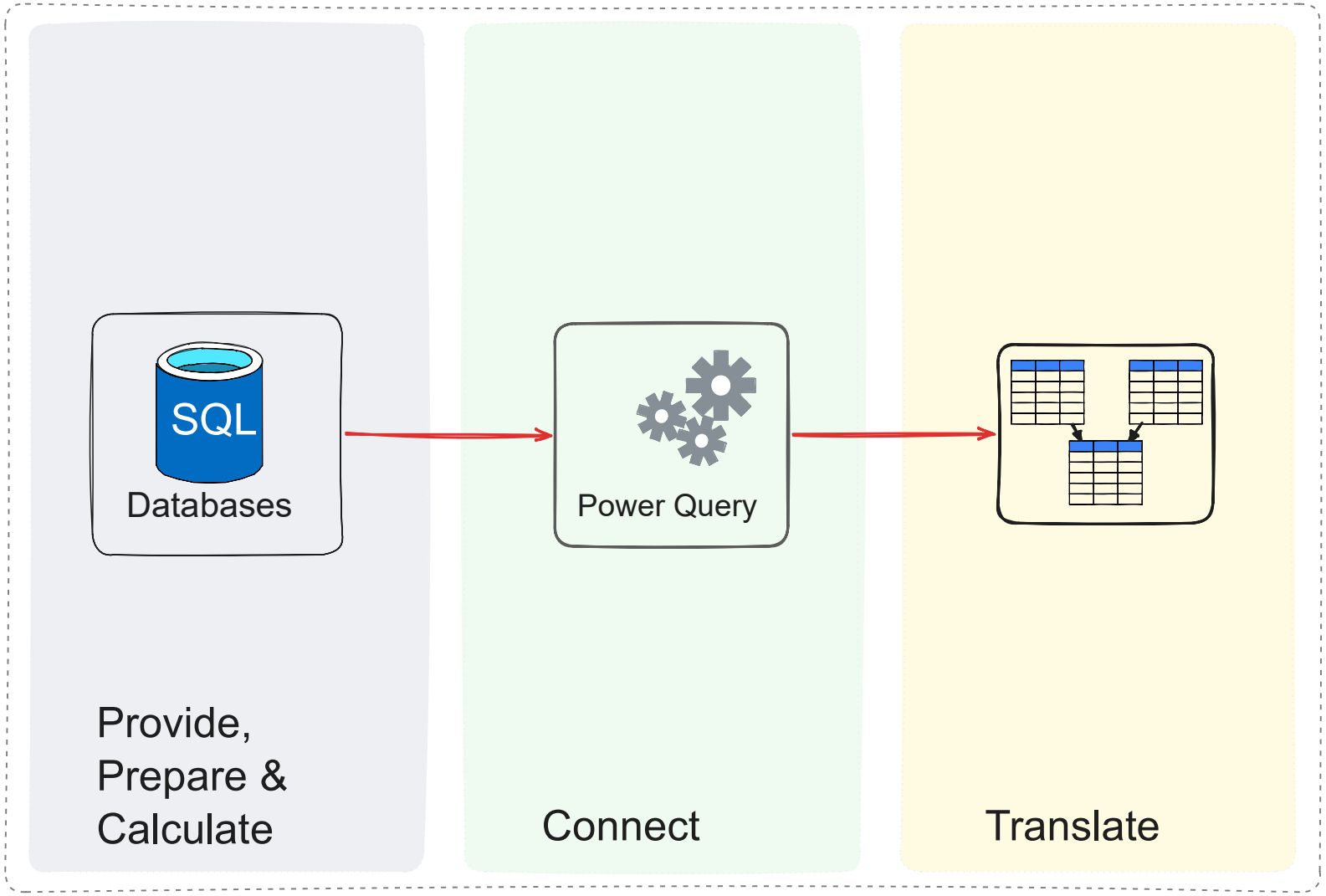


- Supported by all connectors/data sources
- Vertipaq engine
- Decouples import and query workload
- Semantic model security only
- Limited storage & (scheduled) refreshes

DirectQuery Model



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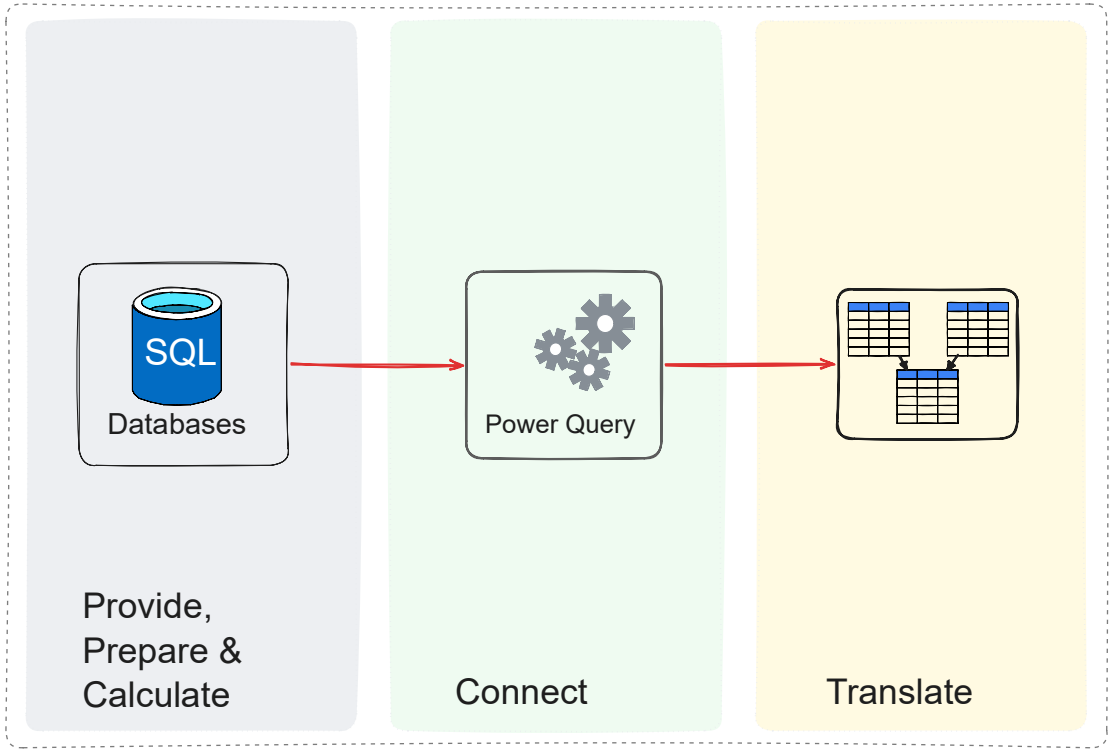
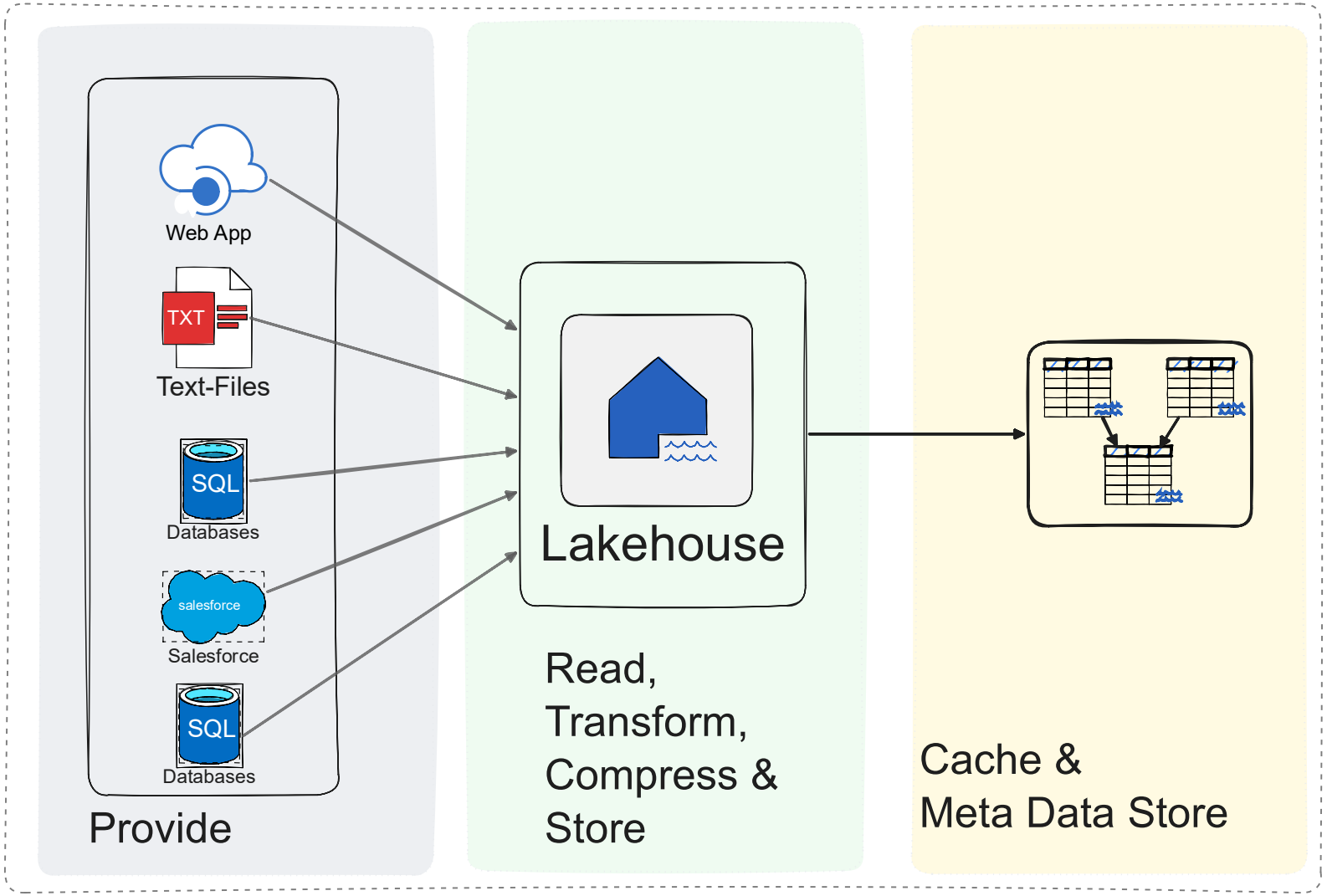


- Data stays at source & always up-to-date
- „no“ model size limits
- Data source security can be leveraged
- Limited DAX & data source support
- Calculated tables not supported

Direct Lake Model



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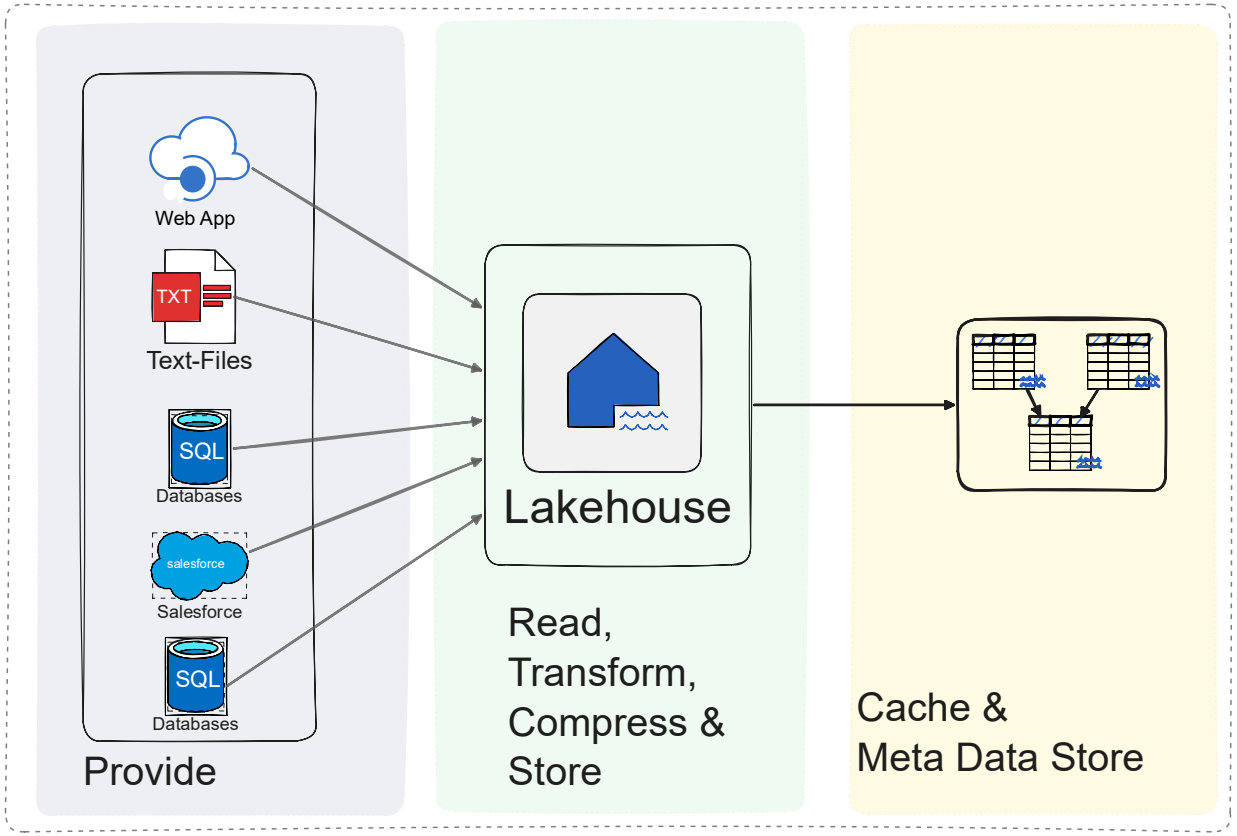
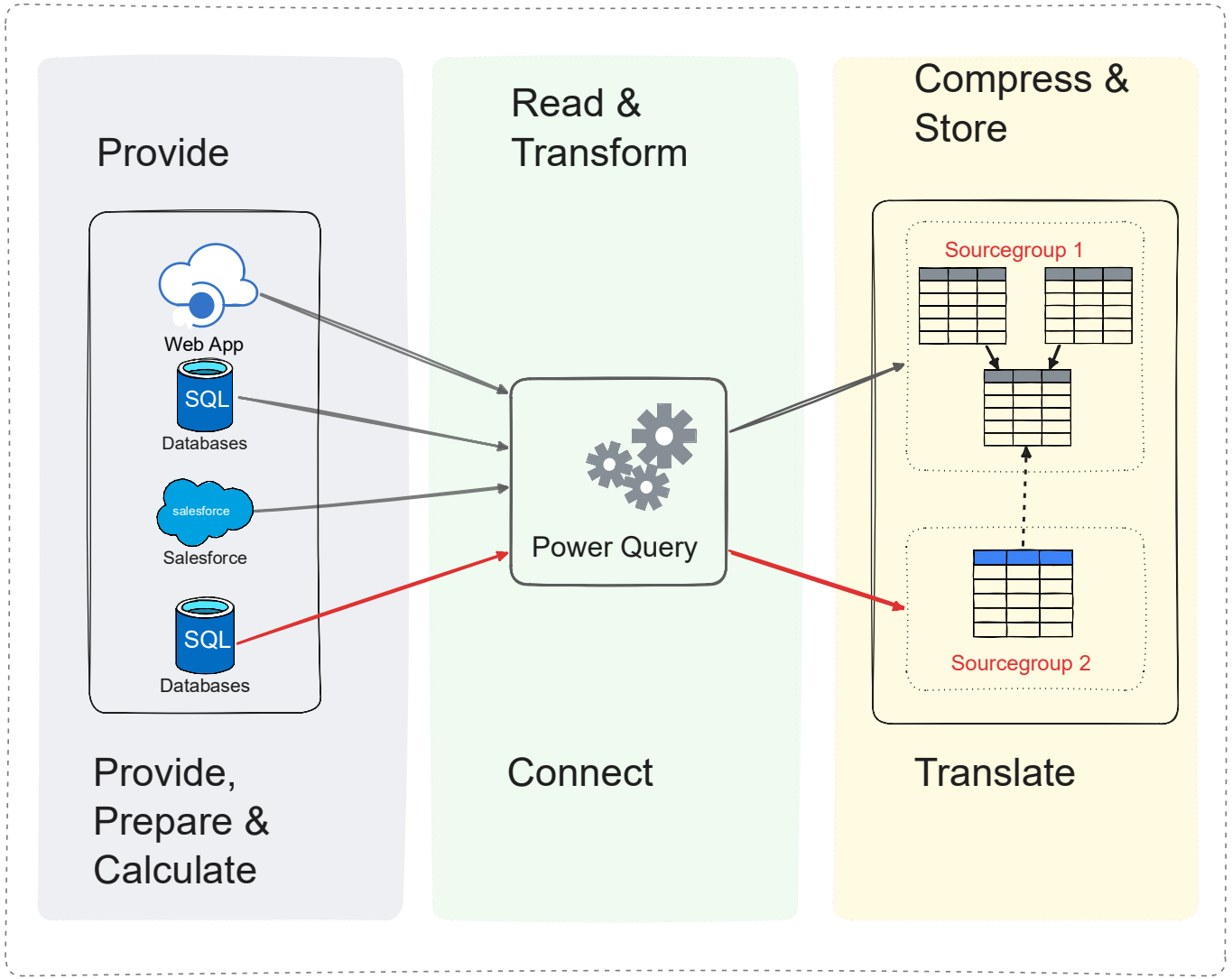


- Vertipaq engine
- Decouples import and query workload
- Model size limited by fabric capacity
- Data source security can be leveraged
- Limited support on calculated table

Composite Models

1. Import & DirectQuery
2. Multi-DirectQuery Mode
3. Import & Direct Lake
4. DirectQuery for Power BI semantic models and Analysis Services

Composite Model – Import + DirectQuery



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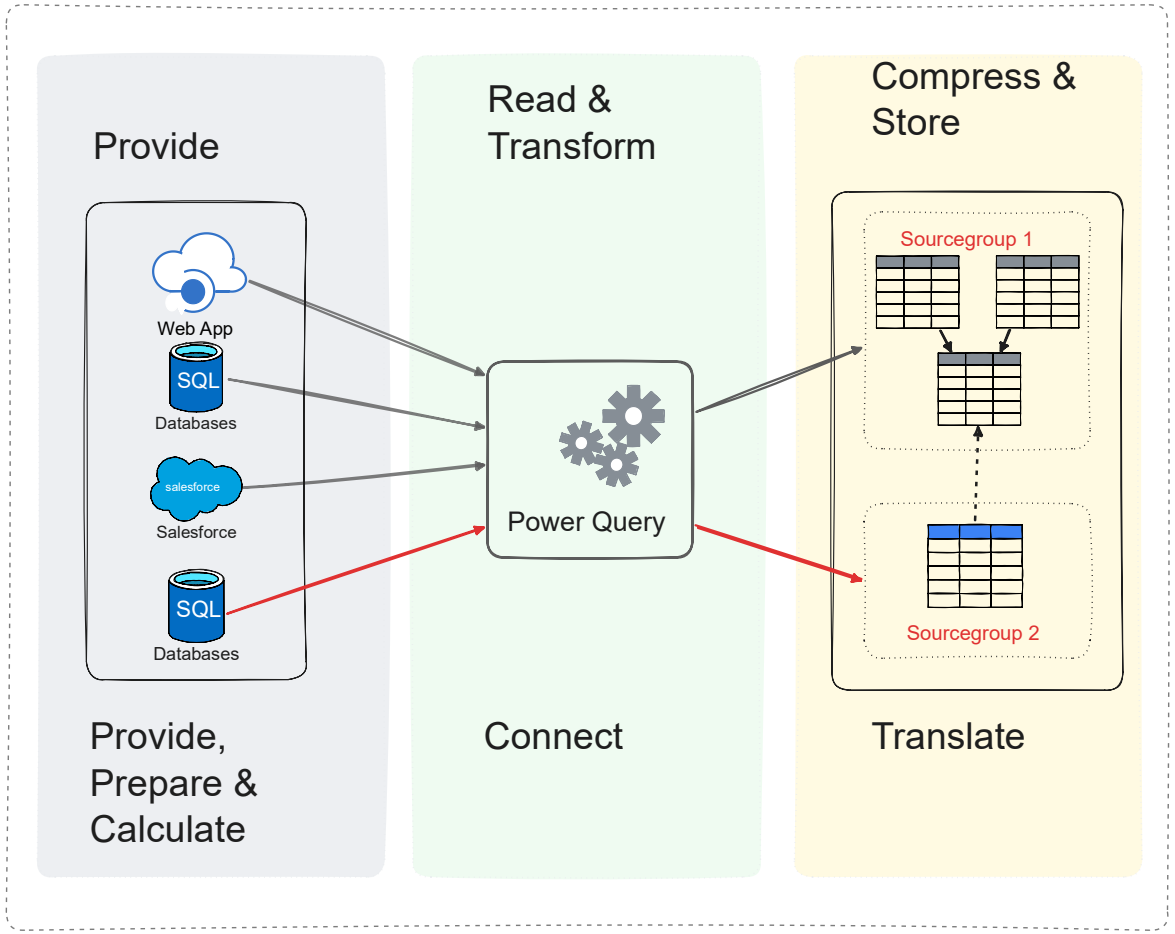
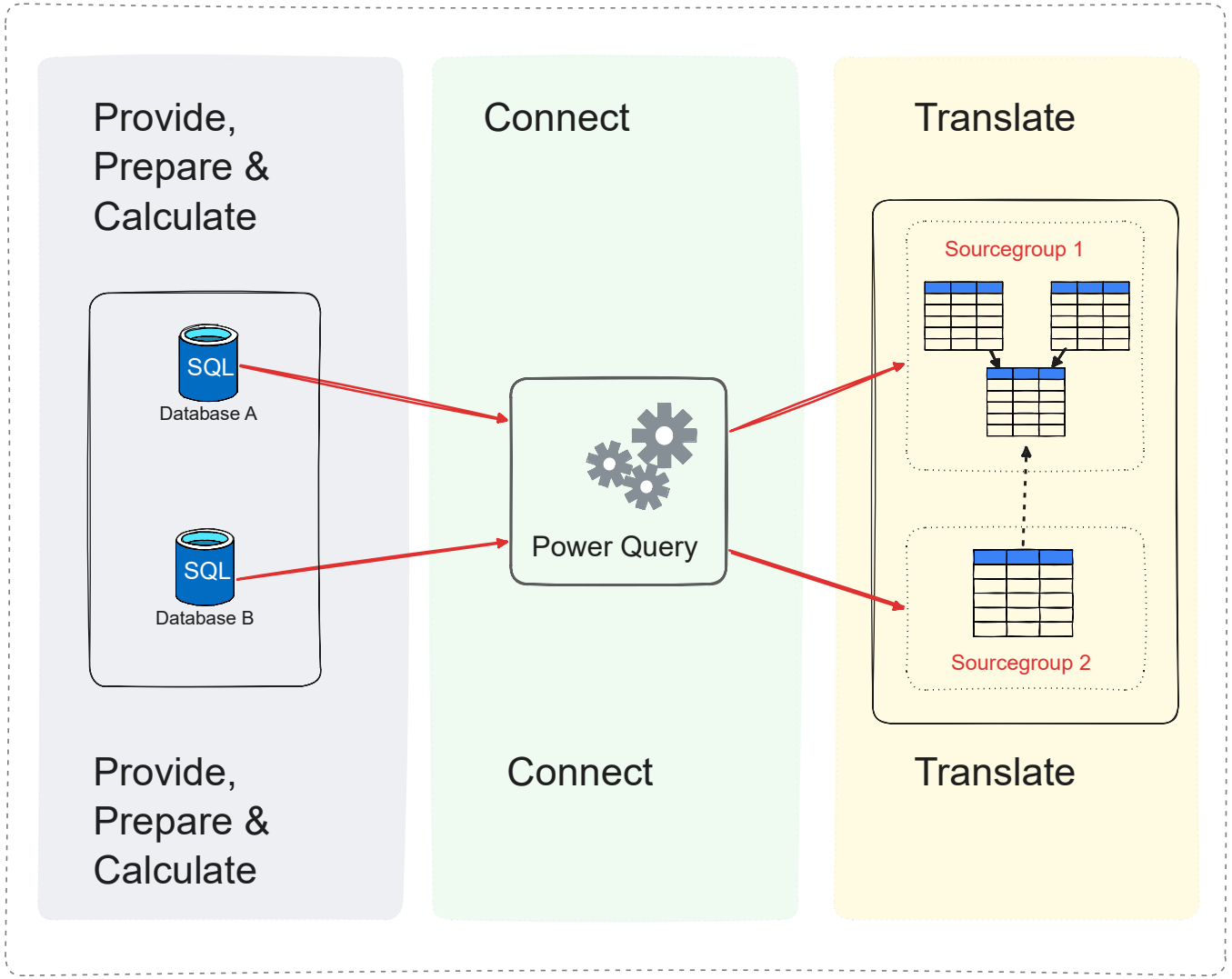


- Import and DirectQuery-Tables
- Dual-Mode possible (regular relationships)
- Limitations of DirectQuery-models apply (DQ only)
- Data source security can be leveraged (DQ only)

Composite Model – DirectQuery + DirectQuery



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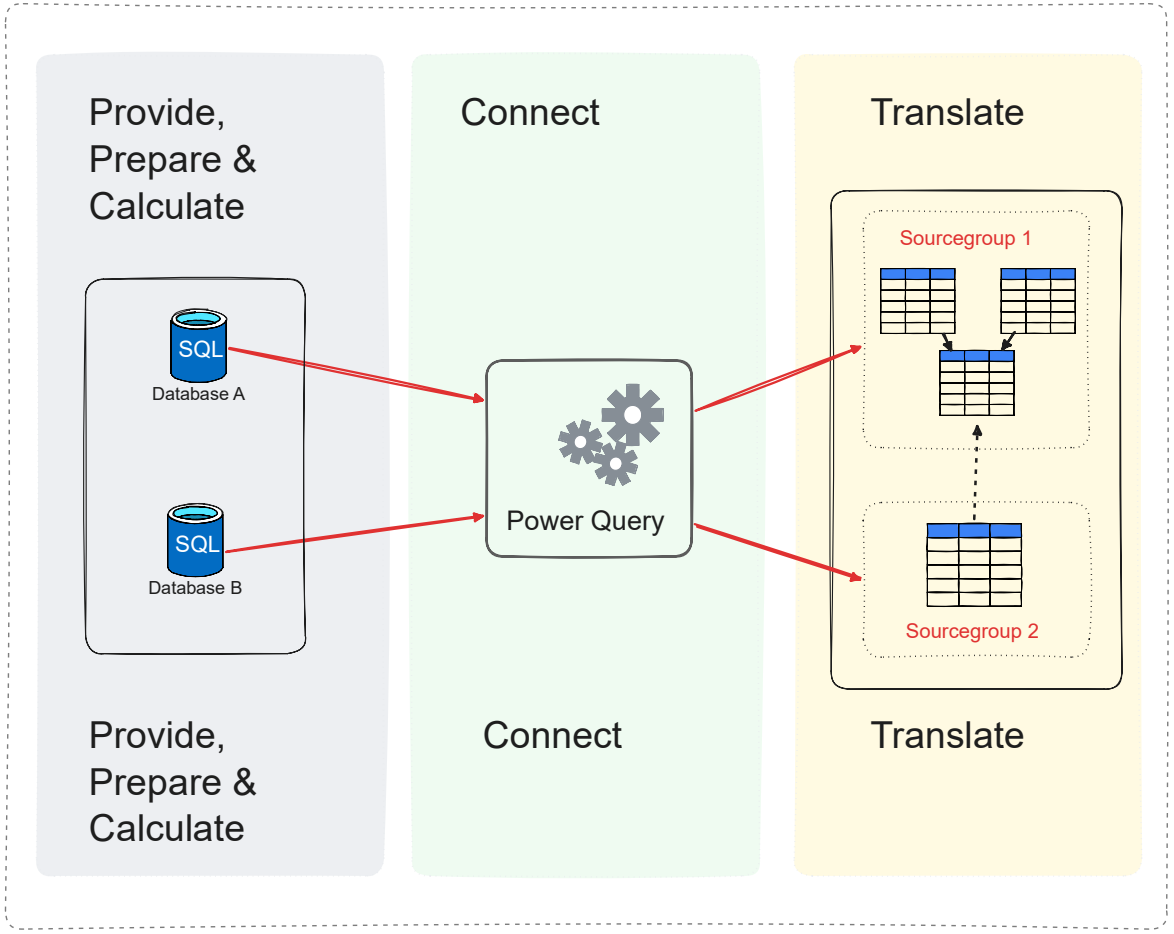
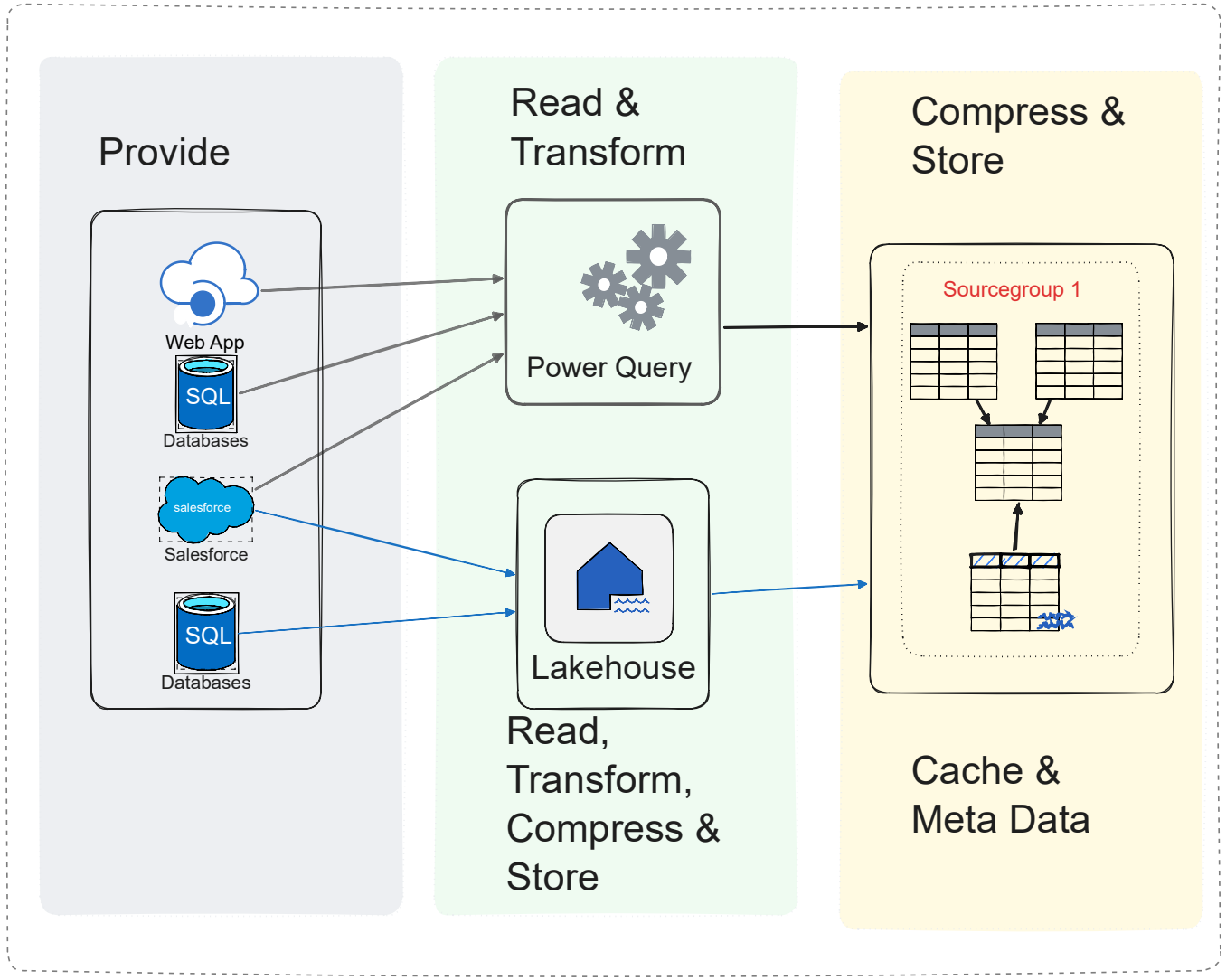


- Tables from different DirectQuery sources
- Limitations of DirectQuery-models apply
- Data source security can be leveraged
- Consider linking data sources upstream

Composite Model – Import + Direct Lake on OneLake



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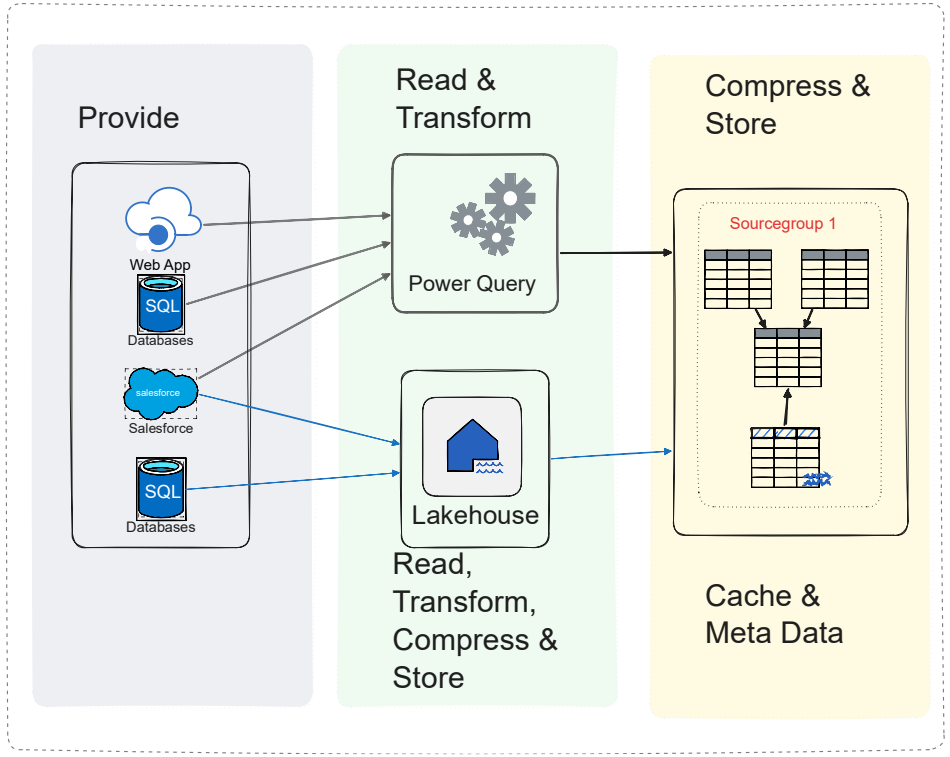
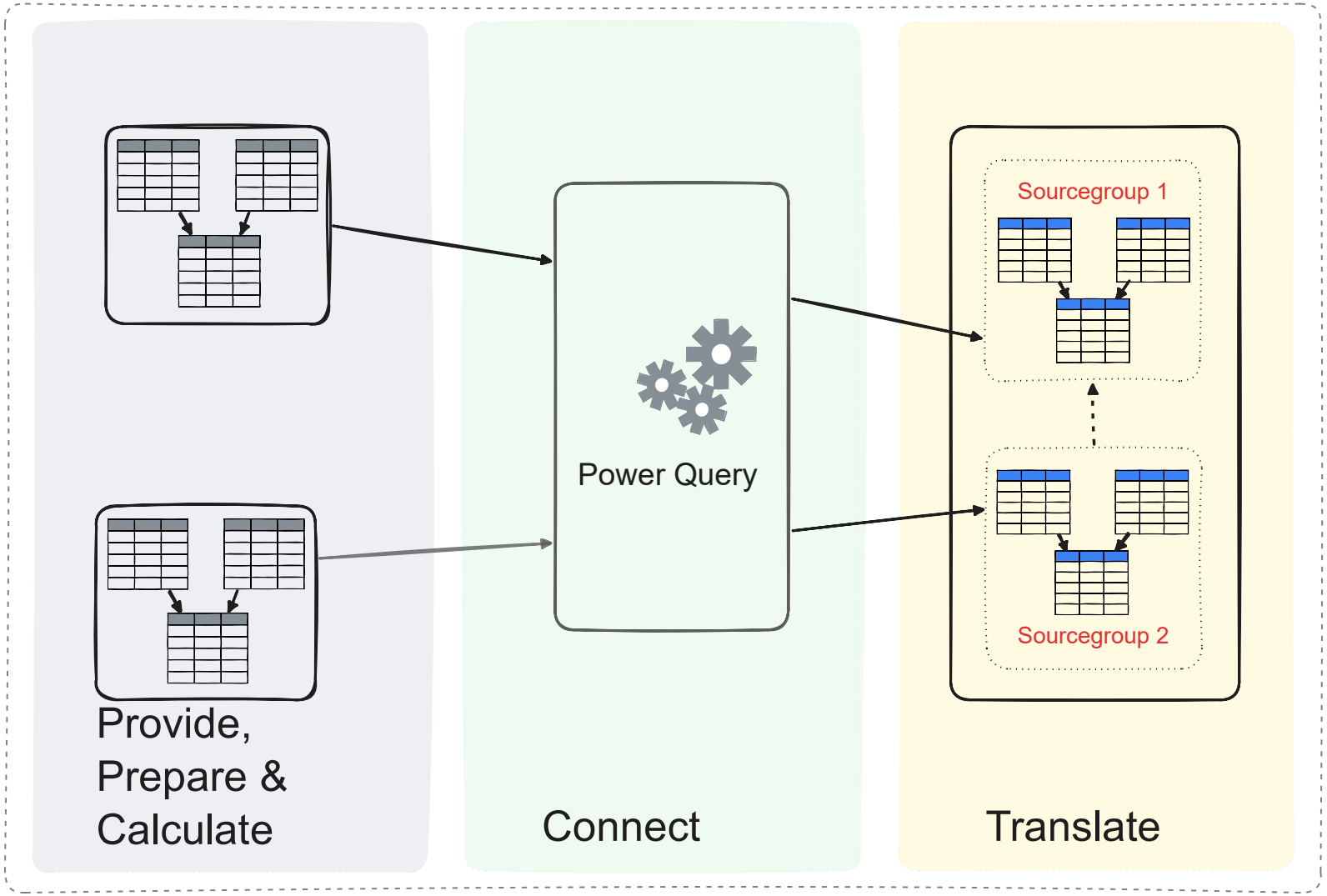


- Contains Import and Direct Lake Tables
- One Sourcegroup
- Improved performance
- All limitations of Direct Lake apply (DL only)

Composite Model – Direct Query for Power BI semantic models

… and Analysis Services 2022+



Composite Model – Direct Query for Power BI semantic models

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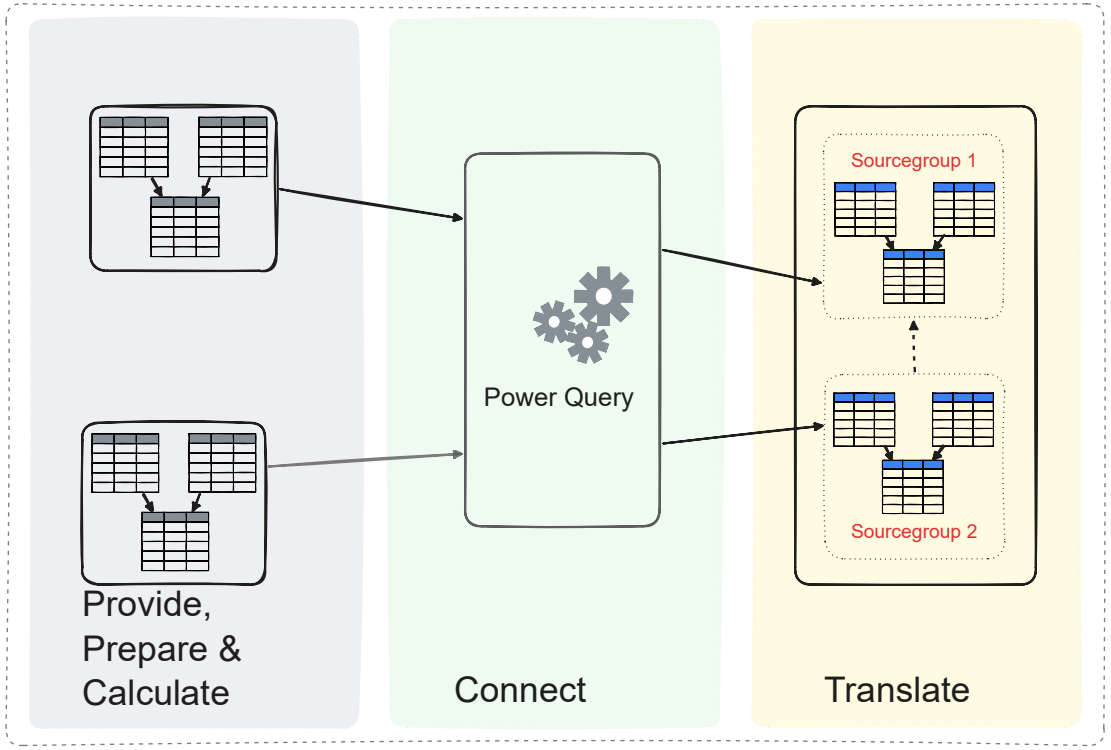

- Multiple semantic models can be combined in one
- Further Data can be added (Import and DQ)
- Source semantic model security is applied
- Consumers need READ-Permisson on all models

Storage Mode Cheat-Sheet

	Importmode	DirectQuery	Direct Lake

Storage Mode Cheat-Sheet

	Importmode	DirectQuery	Direct Lake
Security options	semantic model	data source or semantic model	OneLake, SQL Endpoint or semantic model

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Workspace types	Pro, PPU, Fabric	Pro, PPU, Fabric	Fabric

Which one to use…

- Import Mode
  - Best in query performance, most flexible solution, use it as long as you don't have strong arguments against it
- DirectQuery
  - Best for data freshness on relational or multi-dimensional data sources at the costs of query performance
- Direct Lake
  - Best for data freshness if data resides in OneLake already. Use it for irregular & rarely queried models in Fabric
- Composite Model – Import & DirectQuery
  - Use it for planning/write-back scenarios & if you need aggregations to overcome size limitations in non-fabric environments
- Composite Model – Import & Direct Lake
  - Use it if you need flexibility on dimensions (e.g. calculated columns) combined with huge fact tables
- Composite Model – DirectQuery on PBI
  - Use it for cross model reporting to enhance maintainability or to enable self-service enrichment

Thank you,

stay curious &
please give feedback

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Mastering Power BI Model Types - Import, DirectQuery, or Composite?

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Slides: https://bit.ly/PowerBIModeltypes20260319